

Exercise 3.5 Yaw Assensoh Opoku

1a

The screenshot shows a PostgreSQL client interface with the following components:

- Object Explorer:** Displays a tree view of the database structure. Under 'PostgreSQL 17', the 'Databases (3)' folder is expanded, showing 'ANSWERS 3.2', 'Rockbuster', and 'postgres'. 'ANSWERS 3.2' is selected.
- Query Editor:** Contains the SQL query:

```
1 SELECT title
2 FROM film
3 WHERE title LIKE '%Uptown%'
4
```

A tooltip 'Execute script (F5)' is visible over the 'Execute' button.
- Data Output:** Shows the results of the query in a table format. The table has one column, 'title', with data type 'character varying (255)'. There are four rows of data.
- Footer:** Displays 'Total rows: 4', 'Query complete 00:00:00.130', and 'Ln 3, Col 27'.

	title
1	Chainsaw Uptown
2	Dangerous Uptown
3	Uprising Uptown
4	Uptown Young

1b

Object Explorer

- Servers (4)
 - PostgreSQL 13
 - PostgreSQL 14
 - PostgreSQL 15
 - PostgreSQL 17
 - Databases (3)
 - ANSWERS 3.2
 - Rockbuster
 - postgres
 - Login/Group Roles
 - Tablespaces
 - pgAgent Jobs

Rockbuster/postgres@PostgreSQL 17

Query

```
1 SELECT title, length, rental_rate
2 FROM film
3 WHERE length > 120 AND rental_rate > 2.99
4
```

Data Output

Showing rows: 1 to 157 Page No: 1 of 1

	title character varying (255)	length smallint	rental_rate numeric (4,2)
1	Ali Forever	150	4.99
2	American Circus	129	4.99
3	Dying Maker	168	4.99
4	Baby Hall	153	4.99
5	Casualties Encino	179	4.99
6	Beauty Grease	175	4.99
7	Betrayed Rear	122	4.99
8	Bikini Borrowers	142	4.99
9	Birch Antitrust	162	4.99

Total rows: 157 Query complete 00:00:00.103 LF Ln 3, Col 14

1c

The screenshot shows a database management tool interface with the following components:

- Object Explorer:** A tree view on the left showing the database structure. It includes Servers (4), PostgreSQL 13, PostgreSQL 14, PostgreSQL 15, PostgreSQL 17, Databases (3), ANSWERS 3.2, Rockbuster, postgres, Login/Group Roles, Tablespaces, and pgAgent Jobs.
- Query Editor:** The main area for writing SQL queries. It shows a query for the 'film' table, selecting 'title' and 'rental_duration' where 'rental_duration' is greater than 3 and less than 7.
- Data Output:** A table showing the results of the query. It has two columns: 'title' (character varying (255)) and 'rental_duration' (smallint). The results are displayed in a table with 9 rows.
- Footer:** A status bar at the bottom showing 'Total rows: 606', 'Query complete 00:00:00.111', and 'LF Ln 3, Col 23'.

Query:

```
1 SELECT title, rental_duration
2 FROM film
3 WHERE rental_duration > 3 AND rental_duration < 7;
4
```

Data Output:

	title character varying (255)	rental_duration smallint
1	Grosse Wonderful	5
2	Airport Pollock	6
3	Bright Encounters	4
4	Academy Dinosaur	6
5	Affair Prejudice	5
6	African Egg	6
7	Airplane Sierra	6
8	Aladdin Calendar	6
9	Alamo Videotape	6

Total rows: 606 Query complete 00:00:00.111 LF Ln 3, Col 23

1d

The screenshot shows a database management tool interface. On the left is the 'Object Explorer' pane with a tree view of servers and databases. The main area is divided into a query editor and a results pane. The query editor shows a SQL query: `SELECT title, replacement_cost FROM film WHERE replacement_cost < 14.99`. The results pane shows a table with two columns: 'title' and 'replacement_cost'. The table contains 9 rows of data. The status bar at the bottom indicates 'Total rows: 249', 'Query complete 00:00:00.115', and the current cursor position 'LF Ln 3, Col 31'.

Object Explorer

- Servers (4)
 - PostgreSQL 13
 - PostgreSQL 14
 - PostgreSQL 15
 - PostgreSQL 17
 - Databases (3)
 - ANSWERS 3.2
 - Rockbuster
 - postgres
 - Login/Group Roles
 - Tablespaces
 - pgAgent Jobs

Rockbuster/postgres@PostgreSQL 17*

Query

```
1 SELECT title, replacement_cost
2 FROM film
3 WHERE replacement_cost < 14.99
4
```

Data Output

Showing rows: 1 to 249 Page No: 1 of 1

	title character varying (255)	replacement_cost numeric (5,2)
1	Bright Encounters	12.99
2	Ace Goldfinger	12.99
3	Alien Center	10.99
4	Amistad Midsummer	10.99
5	Anaconda Confessions	9.99
6	Anonymous Human	12.99
7	Antitrust Tomatoes	11.99
8	Apocalypse Flamingos	11.99
9	Argonauts Town	12.99

Total rows: 249 Query complete 00:00:00.115 LF Ln 3, Col 31

1e

The screenshot shows a PostgreSQL management tool interface. On the left is the 'Object Explorer' pane with a tree view of servers and databases. The main area is divided into a query editor and a results pane. The query editor shows a SQL query to select ratings from a 'film' table. The results pane shows the output of the query as a table with two columns: 'rating' and 'mpaa_rating'.

Object Explorer

- Servers (4)
 - PostgreSQL 13
 - PostgreSQL 14
 - PostgreSQL 15
 - PostgreSQL 17
 - Databases (3)
 - ANSWERS 3.2
 - Rockbuster
 - postgres
 - Login/Group Roles
 - Tablespaces
 - pgAgent Jobs

Query Editor

Rockbuster/postgres@PostgreSQL 17

Query

```
1 SELECT rating
2 FROM film
3 WHERE rating = 'PG' OR rating = 'G'
4
```

Data Output

Showing rows: 1 to 372 Page No: 1 of 1

	rating	mpaa_rating
1	PG	
2	G	
3	G	
4	G	
5	PG	
6	G	
7	PG	
8	PG	
9	PG	

Total rows: 372 Query complete 00:00:00.121 LF Ln 3, Col 35

3.

Object Explorer

- Servers (4)
 - PostgreSQL 13
 - PostgreSQL 14
 - PostgreSQL 15
 - PostgreSQL 17
 - Databases (3)
 - ANSWERS 3.2
 - Rockbuster
 - postgres
 - Login/Group Roles
 - Tablespaces
 - pgAgent Jobs

Rockbuster/postgres@PostgreSQL 17

Query

```
1 SELECT
2 COUNT (film_id) AS count_of_movies,
3 AVG (rental_rate) AS average_rental_movies,
4 MAX (rental_duration) AS maximal_renatl_duration,
5 MIN (rental_duration) AS minimum_rental_duration
6 FROM film
7 WHERE rating = 'PG' OR rating = 'G'
8
```

Data Output

Showing rows: 1 to 1 Page No: 1 of 1

	count_of_movies bigint	average_rental_movies numeric	maximal_renatl_duration smallint	minimum_rental_duration smallint
1	372	2.9738709677419355	7	3

Total rows: 1 Query complete 00:00:00.232 LF Ln 1, Co

4

The screenshot displays a database management tool interface. On the left, the 'Object Explorer' pane shows a tree structure of servers and databases. The 'Servers (4)' folder is expanded, showing PostgreSQL 13, 14, 15, and 17. Under PostgreSQL 17, the 'Databases (3)' folder is expanded, showing 'ANSWERS 3.2', 'Rockbuster', and 'postgres'. The 'ANSWERS 3.2' database is selected.

The main window shows a SQL query editor with the following query:

```
1 SELECT rating,  
2 COUNT (film_id) AS count_of_movies,  
3 AVG (rental_rate) AS average_rental_movies,  
4 MAX (rental_duration) AS maximal_renatl_duration,  
5 MIN (rental_duration) AS minimum_rental_duration  
6 FROM film  
7 WHERE rating = 'PG' OR rating = 'G'  
8 GROUP BY rating  
9  
10
```

Below the query editor, the 'Data Output' pane shows the results of the query. The results are displayed in a table with 6 columns: rating, count_of_movies, average_rental_movies, maximal_renatl_duration, and minimum_rental_duration. The table shows 2 rows of data.

	rating mpaa_rating	count_of_movies bigint	average_rental_movies numeric	maximal_renatl_duration smallint	minimum_rental_du smallint
1	PG	194	3.0518556701030928	7	
2	G	178	2.8888764044943820	7	

At the bottom of the interface, a status bar shows 'Total rows: 2', 'Query complete 00:00:00.169', and 'Ln 7, Col 35'.